

BudgetBridge - Milestone 02 Database Design Package

CP476B Group 6 - Personal Expense Tracker

1. Database Design Overview

BudgetBridge uses a normalized relational database design. The main workflow is centered on users, categories, and transactions. Additional tables support sessions, budget limits, and activity logs for future Milestone 03 integration. The schema uses primary keys, foreign keys, constraints, indexes, and sensible MySQL data types.

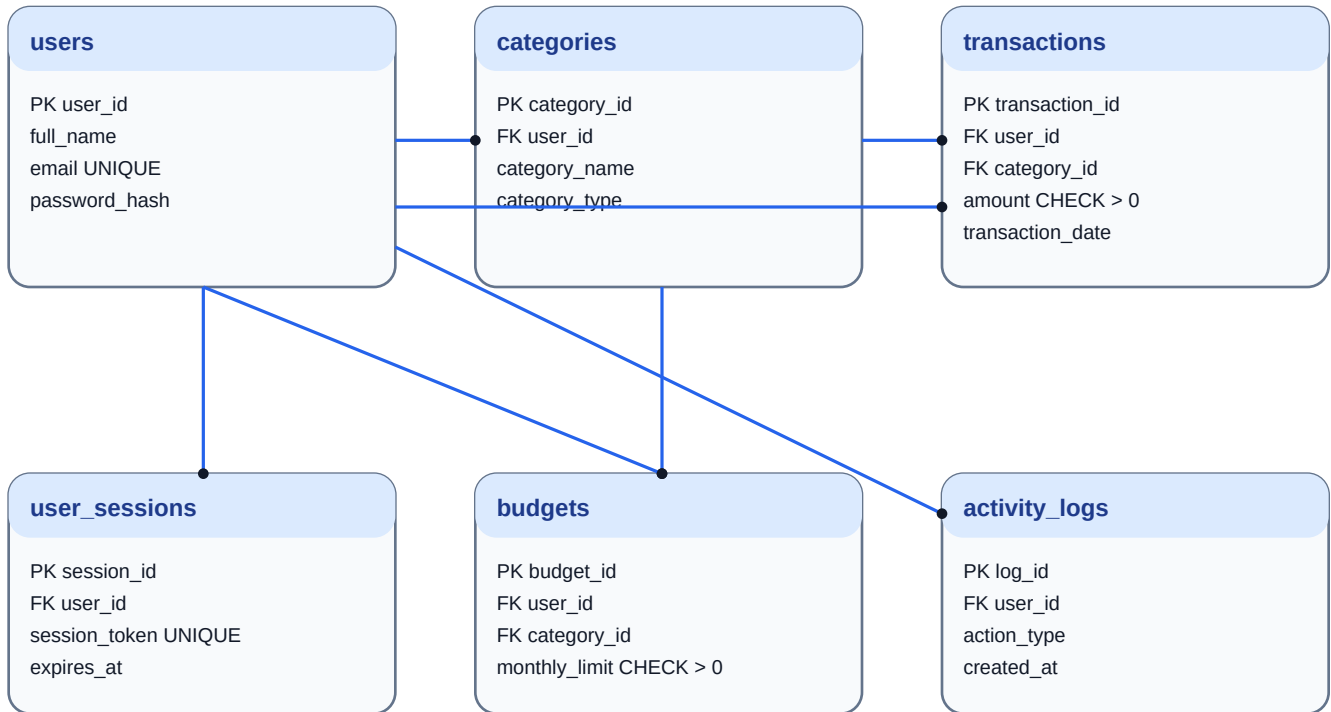
Table	Purpose	Key Fields
users	Stores registered user accounts.	user_id, full_name, email, password_hash
categories	Stores default and user-created income/expense categories.	category_id, user_id, category_name, category_type
transactions	Stores income and expense records.	transaction_id, user_id, category_id, amount, transaction_date
user_sessions	Supports future login session management.	session_id, user_id, session_token, expires_at
budgets	Supports future monthly category budget limits.	budget_id, user_id, category_id, monthly_limit
activity_logs	Stores important user actions for auditing/debugging.	log_id, user_id, action_type, action_description

2. High-Level Relationships

- One user can have many transactions; each transaction belongs to one user.
- One user can have many categories; each custom category belongs to one user.
- One category can be used by many transactions.
- One user can have many sessions.
- One user can have many budget records; each budget record belongs to one category.
- One user can have many activity log records.

3. Relational Diagram

The following diagram shows the main table relationships. Primary keys are marked with PK and foreign keys are marked with FK.



Relationship style: one-to-many from users to related records; categories connect to transactions and budgets.

4. Normalization and Constraint Notes

Design Choice	Reason
Separate users table	Keeps account information independent from transactions and supports one user owning many records.
Separate categories table	Avoids repeating category names directly inside every transaction.
transactions references users and categories	Supports clear ownership and category-based summaries.
CHECK constraints for amount/monthly_limit	Prevents invalid negative or zero values.
UNIQUE email	Prevents duplicate accounts with the same email.
Indexes on transaction date/category/type	Improves filtering and report queries.

5. SQL CREATE TABLE Statements

The following SQL is also provided separately in database/schema.sql.

```
-- BudgetBridge Personal Expense Tracker
-- CP476B Milestone 02 Database Schema
-- Target DBMS: MySQL 8.x

CREATE DATABASE IF NOT EXISTS budgetbridge_db;
USE budgetbridge_db;

CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

CREATE TABLE categories (
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NULL,
    category_name VARCHAR(80) NOT NULL,
    category_type ENUM('income', 'expense') NOT NULL,
    is_default BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_categories_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE CASCADE,
    CONSTRAINT uq_category_per_user
        UNIQUE (user_id, category_name)
);

CREATE TABLE transactions (
    transaction_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    category_id INT NOT NULL,
    transaction_type ENUM('income', 'expense') NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    transaction_date DATE NOT NULL,
    description VARCHAR(255),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    CONSTRAINT chk_transaction_amount_positive CHECK (amount > 0),
    CONSTRAINT fk_transactions_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_transactions_category
        FOREIGN KEY (category_id) REFERENCES categories(category_id)
        ON DELETE RESTRICT
);

CREATE TABLE user_sessions (
    session_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    session_token VARCHAR(255) NOT NULL UNIQUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    expires_at TIMESTAMP NOT NULL,
    CONSTRAINT fk_sessions_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE CASCADE
);
```

```

);

CREATE TABLE budgets (
    budget_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    category_id INT NOT NULL,
    monthly_limit DECIMAL(10,2) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_budget_limit_positive CHECK (monthly_limit > 0),
    CONSTRAINT fk_budgets_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_budgets_category
        FOREIGN KEY (category_id) REFERENCES categories(category_id)
        ON DELETE CASCADE,
    CONSTRAINT uq_budget_user_category
        UNIQUE (user_id, category_id)
);

CREATE TABLE activity_logs (
    log_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    action_type VARCHAR(50) NOT NULL,
    action_description VARCHAR(255) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_logs_user
        FOREIGN KEY (user_id) REFERENCES users(user_id)
        ON DELETE CASCADE
);

CREATE INDEX idx_transactions_user_date ON transactions(user_id, transaction_date);
CREATE INDEX idx_transactions_user_category ON transactions(user_id, category_id);
CREATE INDEX idx_transactions_type ON transactions(transaction_type);
CREATE INDEX idx_categories_type ON categories(category_type);

-- Default categories can be inserted for each new user by server-side application logic.
-- Example default categories: Income, Food, Rent, Transportation, School, Entertainment, Other.

```